

第四章 衍射

§ 1

1.2 惠更斯-菲涅耳原理

波前上任意一面元 $d\vec{z}$ 都可以看作振动的中心,

空间中 P 点的振动是它们发出振动的相干叠加

$$\tilde{u}(P) = \oint d\tilde{u}(P)$$

球面波

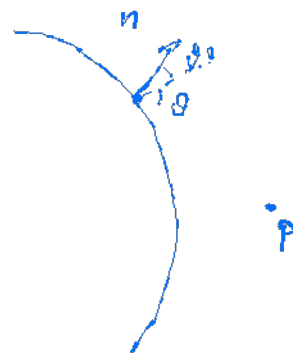
倾斜因子

假设 $d\tilde{u}(P) \propto d\vec{z} \cdot \tilde{u}_0(\Omega) \cdot \frac{e^{ikr}}{r} \cdot F(\theta, \theta)$

$$= \frac{1}{2} (\cos \theta_0 + \cos \theta)$$

→ 菲涅耳衍射积分: 瞳函数

$$\begin{aligned} u(P) &= k \cdot \oint \tilde{u}_0(\Omega) F(\theta, \theta) \frac{e^{ikr}}{r} d\vec{z} \\ &= \frac{-i}{\lambda} \cdot \frac{e^{ikr}}{\lambda} \end{aligned}$$



基尔霍夫边界条件:

取波前 Σ : Σ_0 光孔部分

Σ_1 光屏部分

Σ_2 半径为 ∞ 的半球面.

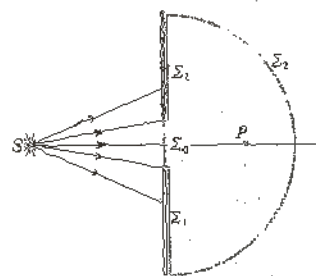


图 4-7 关于基尔霍夫边界条件的说明

$$\tilde{u}(P) = \frac{-i}{2\lambda} \iint_{\Sigma_0} (\cos \theta_0 + \cos \theta) \tilde{u}_0(\Omega) \cdot \frac{e^{ikr}}{r} d\vec{z}$$

↓ (傍轴条件: $r \approx r_0$ 且 $\theta = \cos \theta_0 \approx 1$)

$$\tilde{u}(P) = \frac{-i}{\lambda r_0} \iint \tilde{u}_0(\Omega) e^{ikr} d\vec{z}$$

1.3 巴比涅原理

$$\iint_{\Sigma_0} + \iint_{\Sigma_1} = \iint_{\Sigma_0} \rightarrow \tilde{u}_0(P) + \tilde{u}_1(P) = \tilde{u}_0(P)$$

对点光源照明, 衍光衍成像系统
在像平面上接收衍射图样的系统

$$\tilde{u}_0(P) = -\tilde{u}_1(P) \text{ (除像点)}$$

§ 2. 菲涅耳衍射和圆屏衍射

2.2 半波带法

各半波带在P点相位差 π

$$A_1(P) = A_1(P) e^{i\varphi}$$

$$A_2(P) = A_2(P) e^{i\varphi + \pi}$$

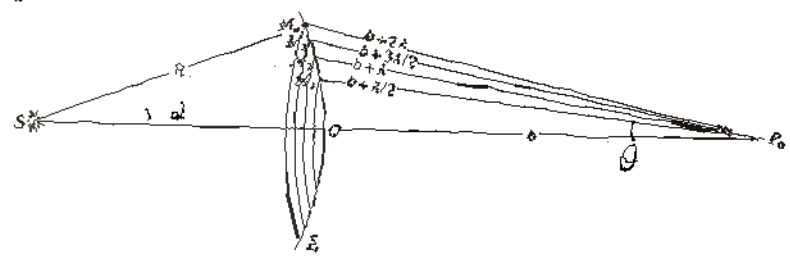


图 4-11 半波带法

$$A(P) = | \sum_{k=0}^{\infty} \tilde{A}_k(P) | = A_1(P) - A_2(P) + \dots + (-1)^{n-1} A_n(P)$$

$$\text{而 } A_k(P) \propto F(\theta_0, \theta) \cdot \frac{d\tilde{E}_k}{r_k}$$

$$F(\theta_0, \theta) = \frac{1}{2} (\cos \theta_0 + \cos \theta)$$

$$d\tilde{E}_k = 2\pi R \sin \alpha R d\alpha = -2\pi R^2 \alpha (\cos \alpha)$$

$$r_k = \sqrt{R^2 + (R+b)^2} - 2R(R+b) \cos \alpha$$

$$\frac{d\tilde{E}_k}{r_k} = -2\pi R^2 \cdot \frac{d \left(\frac{R^2 + (R+b)^2 - r_k^2}{2R(R+b)} \right)}{r_k}$$

$$= 2\pi R^2 \cdot \frac{2r_k dr}{r_k \cdot 2R(R+b)} = \frac{2\pi R dr}{R+b}$$

$$dr = r_{k+1} - r_k = \frac{\lambda}{2} \Rightarrow \frac{d\tilde{E}_k}{r_k} = \frac{\pi R \lambda}{R+b}, \quad A_k(P) \propto \frac{\cos \theta + \cos \theta_0}{2}$$

$$A(P) = A_1(P) - A_2(P) + A_3(P) - \dots$$

$$A(P) = \frac{1}{2} A_1(P) = \frac{1}{2} A(P_0)$$

⇒ 角由传播时波前在P处振幅为第一个半波带的1/2。

① 同孔衍射:

圆孔包含 1 个 $\rightarrow A(P) = A_0(P)$ 亮

半波带: 2 个 $\rightarrow A(P) = A_1(P) + A_2(P) \approx 0$ 暗

奇数 $\rightarrow$ 亮
偶数 $\rightarrow$ 暗

改变 $b \rightarrow$ 改变 $\varphi \rightarrow A(P)$ 相位变化

② 用屏: 遮住前 k 个半波带时

$$A(P) = A_{2k+1}(P) + \dots + (-1)^{n-k} A_n(P) \\ = \frac{1}{2} A_{2k+1}(P) \quad \text{总是亮纹}$$

2.3 矢量图法 (如果圆孔不包含一个完整的半波带)

在一个半波带中, 划分出 m 个部分

$$b + \frac{\lambda}{2m}, \quad \dots, \quad b + \frac{(m-1)\lambda}{2m}, \quad b + \frac{\lambda}{2} \\ \downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow \\ \triangle A_1 \qquad \qquad \qquad \triangle A_{m-1} \qquad \qquad \qquad \triangle A_m$$

$$\delta = \frac{2\pi}{\lambda} \cdot \frac{1}{2m} \cdot \lambda = \frac{\pi}{m} = \angle A_1 A_2 \dots A_m$$

倾斜因子 $\rightarrow |\triangle A_m|$ 略小于 $|\triangle A_1|$

$$\rightarrow A = \frac{1}{2} A_1$$

例：圆孔衍射，包含半个波带的轴上衍射强度



$$\rightarrow A' = \frac{\sqrt{2}}{2} A = \frac{1}{\sqrt{2}} A$$

$$I' = 2I$$

例：以角内传播为例验证惠-菲原理并求出衍射积分系数

球面波： $u = \frac{G}{r} e^{ikr}$

$r=R$: $u(\theta) = \frac{G}{R} e^{ikR}$

$r=R+b$: $u_1(P) = \frac{G}{R+b} e^{ik(R+b)}$

角内传播的 $u(P) = \frac{1}{2} u_1(P)$

$$u_1(P) = \iint_{\Sigma} K \cdot F(\theta, \phi) \tilde{u}_0(\theta) \frac{e^{ikr}}{r} d\tilde{\Sigma}$$

Σ 内 $F(\theta, \phi) \approx 1$, $\tilde{u}_0(\theta) = u(\theta)$

$$\rightarrow u_1(P) = \frac{aKe^{ikR}}{R} \iint \frac{e^{ikr}}{r} d\tilde{\Sigma} = \frac{aKe^{ikR}}{R} \cdot \iint e^{ikr} \cdot \frac{2\pi Rdr}{R+b}$$

$$= \frac{2\pi aKe^{ikR}}{R+b} \int_b^{b+\frac{\Delta}{2}} e^{ikr} dr = \frac{2\pi aKe^{ikR}}{(R+b)ik} e^{ikb} (e^{ik\frac{\Delta}{2}} - 1)$$

$$= \frac{a \cdot e^{ik(R+b)}}{R+b} \frac{2\pi k(e^{ik\frac{\Delta}{2}} - 1)}{ik} = u(P) = \frac{4\pi k}{ik}$$

$$\rightarrow -\frac{4\pi k}{ik} = 2 \rightarrow K = -\frac{ik}{2\pi} = -\frac{i}{\lambda}$$

7.4 菲涅耳波带片

$$r_k = b + \frac{k\lambda}{2}$$

$$\rho_k = R \sin \alpha_k \approx \sqrt{\frac{Rb}{R+b}} k\lambda = \sqrt{\rho_1} \quad \text{其中 } \rho_1 = \sqrt{\frac{Rb\lambda}{R+b}}$$

记 $\rho = \frac{R\lambda}{2} = \frac{\lambda}{\rho_1^2}$

次焦点 ?

$$\frac{1}{R} + \frac{1}{b} = \frac{1}{f} \quad \rightarrow \text{透镜片云一焦距}$$

虚像云云

§ 3 夫琅禾费衍射和透镜衍射

3.1 单缝衍射

$$\Delta L = a \sin \theta$$

① 矢量图解法

$$\delta = \sum \Delta L = \frac{2\pi C \sin \theta}{\lambda}$$

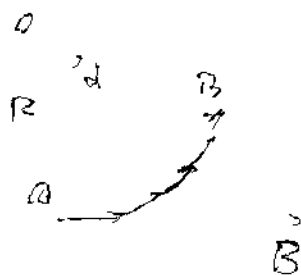
$$2\alpha = \delta$$

$$\overline{AB} = 2R \sin \alpha = R_0$$

$$\widehat{AB} = 2R \alpha = R_0$$

$$\Rightarrow I = I_0 \cdot \frac{\sin^2 \alpha}{\alpha^2} \quad \text{其中 } \alpha = \frac{\pi a \sin \theta}{\lambda}$$

单缝衍射图样



② 复数积分法

★ 经透镜变换后，球面波 $\frac{e^{ikr}}{r} \rightarrow \frac{e^{ikr}}{f}$

$$\tilde{u}(0) = \frac{-i}{\lambda f} \iint \tilde{u}_0 e^{ikr} dx dy$$

$$kr = -\lambda \sin \theta$$

$$= \left(\frac{-i \tilde{u}_0}{\lambda f} \cdot \int dy \right) e^{ikr_0} \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{ikx \sin \theta} dx = C \cdot \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{ikx \sin \theta} dx$$

$$= C \cdot \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{-ikx \sin \theta} dx = \frac{C}{-ik \sin \theta} \left(e^{-\frac{ik a \sin \theta}{2}} - e^{\frac{ik a \sin \theta}{2}} \right)$$

$$= -\frac{C i}{k \sin \theta} 2 i \sin \left(\frac{k a \sin \theta}{2} \right) = a C \cdot \sin \alpha \frac{k a \sin \theta}{2}$$

$$= a \cdot C \cdot \sin \alpha \quad \text{其中 } \alpha = \frac{k a \sin \theta}{2}$$

取 $\theta = 0$, $\alpha = 0$, $\sin \alpha = 0$, 得 $\tilde{u}_0 = a C$

$$\therefore \tilde{U}(\theta) = \tilde{U}_0 \cdot \sin C(\theta)$$

$$I = \tilde{U}(\theta) \tilde{U}^*(\theta) = I_0 \sin^2 \theta$$

3.3 矩形衍射

取波前上点 $Q(x, y)$, 屏上点 $P(x', y')$

衍射角 (透镜光心到 P 点方向) θ_1, θ_2

$$r = -x \sin \theta_1 - y \sin \theta_2$$

$$\tilde{U}(\theta_1, \theta_2) = C \iint e^{ikr} dx dy$$

$$= C \int \left(\int_{-\frac{a}{2}}^{\frac{a}{2}} e^{ik(-x \sin \theta_1 - y \sin \theta_2)} dy \right) dx$$

$$= C \int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{1}{-ik \sin \theta_1} \left(e^{ik(-x \sin \theta_1 - \frac{a}{2} \sin \theta_2)} - e^{ik(-x \sin \theta_1 + \frac{a}{2} \sin \theta_2)} \right) dx$$

$$= C \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{-ikx \sin \theta_1} \left(e^{ik(-\frac{a}{2} \sin \theta_2)} - e^{ik(\frac{a}{2} \sin \theta_2)} \right) dx$$

$$= C \left[e^{ik(-\frac{a}{2} \sin \theta_1)} - e^{ik(\frac{a}{2} \sin \theta_1)} \right] \left[e^{ik(-\frac{a}{2} \sin \theta_2)} - e^{ik(\frac{a}{2} \sin \theta_2)} \right]$$

$$= C \frac{[-ik \sin \theta_1] [-ik \sin \theta_2]}{k^2 \sin \theta_1 \sin \theta_2}$$

$$= C \sin C(\alpha_1) \sin C(\alpha_2) \quad \text{其中 } \alpha_1 = \frac{ka \sin \theta_1}{2} \quad \alpha_2 = \frac{ka \sin \theta_2}{2}$$

$$I = I_0 \sin^2 C(\alpha_1) \sin^2 C(\alpha_2)$$

3.4 单缝衍射图样的特点

(1) 主极大

$$\alpha = 0$$

(2) 次极大

$$\frac{d}{d\alpha} (\sin C \alpha) = 0 \rightarrow \alpha = \tan \alpha \quad \text{解得 } \alpha = \pm 1.43\pi, \dots$$

(3) 暗纹位置

$$\alpha = \pm k\pi \quad (k \in \mathbb{N}) \quad \text{即 } \frac{ka \sin \theta}{\lambda} = k\pi \quad \sin \theta = \pm \frac{k\lambda}{a}$$

4) 角宽度

傍轴条件: $\Delta\theta \approx \frac{\lambda}{a}$

平面波自由传播 $\Delta\theta \rightarrow 0$
 $\lambda \ll$ 短波 $\Delta\theta \rightarrow 0 \Rightarrow$ 几何光学

例 $\lambda = 0.6 \mu\text{m}$ $a = 20 \mu\text{m}$ $f = 20 \text{ cm}$ 求零级斑的线宽和半角宽

$$\Delta\theta = \frac{\lambda}{a} \quad \Delta s = 2f \Delta\theta$$

几何光学极限下的夫琅禾费单缝衍射

$$\tilde{u}(p) = aC \cdot \text{sinc} \alpha \quad \text{其中 } \alpha = \frac{\pi a \sin \theta}{\lambda} \approx \frac{\pi a \theta}{\lambda} \quad C \propto \frac{1}{\lambda} \left(\frac{-i}{\lambda f} \right)$$

$$\tilde{u}(p) \propto \frac{a}{\lambda} \frac{\sin(\frac{\pi a \theta}{\lambda})}{\frac{\pi a \theta}{\lambda}} = \frac{\sin(\frac{\pi a \theta}{\lambda})}{\pi \theta} = \frac{1}{\lambda} \frac{\sin(p\theta)}{\theta} \quad \text{其中 } p = \frac{\pi a}{\lambda}$$

$$\Rightarrow \delta(\theta, p) = \frac{1}{\pi} \frac{\sin(p\theta)}{\theta}$$

$$\textcircled{1} \int_{-\infty}^{\infty} \delta(\theta, p) d\theta = 1$$

$$\textcircled{2} \delta(\theta) = \lim_{p \rightarrow \infty} \delta(\theta, p) = \frac{1}{\pi} \lim_{p \rightarrow \infty} \frac{\sin(p\theta)}{\theta}$$

$$\int_{-\infty}^{+\infty} \delta(\theta) a\theta = 1$$

$$\text{对在 } \theta=0 \text{ 处连续函数 } f(\theta) \quad \int_{-\infty}^{+\infty} \delta(\theta) f(\theta) a\theta = f(0)$$

§ 4 光学仪器的像分辨率

4.1 夫琅禾费圆孔衍射

引入一阶贝塞尔函数 J_1 ,

$$\tilde{U(\theta)} \propto \frac{J_1(x)}{x} \quad \text{其中 } x = \frac{2\pi a}{\lambda} \sin \theta$$

$$I \propto [J_1(x)]^2$$

$$\text{主极大角半径: } \Delta \theta = 0.61 \frac{\lambda}{D} = 1.72 \frac{\lambda}{D}$$

人的瞳孔艾里斑大小? 10^{-5} m .

4.2 望远镜的分辨率

$$\text{最佳视场放大率 } M = \frac{\delta \theta_e}{\delta \theta_{\min}} \quad \text{其中 } \delta \theta_e \text{ 是人眼分辨率, 一般为 } 1'$$

$\delta \theta_{\min}$ 是望远镜分辨率

4.3 球面波照明条件下衍射

由 L-H 定理

$$n D \sin \theta = n' D' \sin \theta' = n'' D'' \sin \theta'' = 1.22 \lambda$$

4.4 显微镜的分辨率

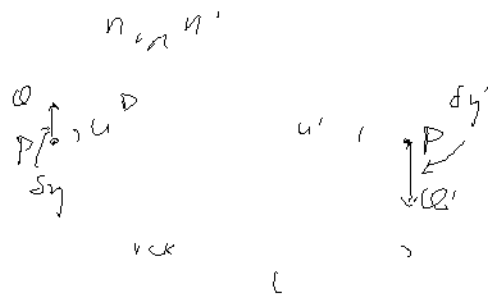
$$\text{像的艾里斑: } \Delta \theta = \frac{1.22 \lambda}{n' D}$$

$$\text{Abel 正弦条件: } n \sin u \cdot \delta y = n' \sin u' \cdot \delta y'$$

$$\sin u \approx \frac{D}{2L}$$

$$\delta y' \geq \Delta \theta \Rightarrow \delta y_{\min} = \frac{0.61 \lambda}{N.A.} \quad \text{数值孔径 } N.A. = n \sin u$$

减小 $\delta y_{\min}$: ① 增加 N.A. : 油浸物镜



② 衍射光强分布

§ 5 多缝夫琅禾费衍射和光栅

5.1 衍射图样

- ① 主极大在正前方
- ② $N \uparrow$ 主极大宽度小
- ③ 相邻主极大之间有 $N-1$ 条暗纹
 $N-2$ 个次极大

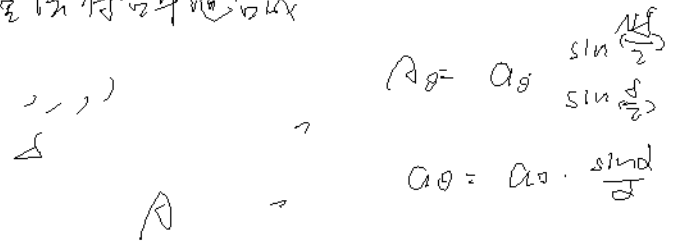
5.2 强度分布

单缝衍射时 $I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$ 其中 $\alpha = \frac{\pi a}{\lambda} \sin \theta$, a 为缝宽

对于 θ 方向光线

$$\delta = \frac{2\pi}{\lambda} \cdot (a+b) \sin \theta$$

用矢量法将各单缝合成



$$I_\theta = A_\theta^2 = I_0 \cdot \left(\frac{\sin \alpha}{\alpha} \right)^2 \cdot \left(\frac{\sin N\beta}{\sin \beta} \right)^2$$

单缝衍射因子 缝间干涉因子

其中 $\alpha = \frac{\pi a}{\lambda} \sin \theta$ $\beta = \frac{\pi (a+b)}{\lambda} \sin \theta = \frac{\pi d}{\lambda} \sin \theta$

5.3 缝间干涉因子 $\left(\frac{\sin N\beta}{\sin \beta} \right)$

(1) 主极大 $\beta = k\pi$, $\frac{\sin N\beta}{\sin \beta} = N$

$\sin \theta_k = \frac{k\lambda}{a} \rightarrow \begin{cases} \text{最大级数 } k_{\max} = \lfloor \frac{a}{\lambda} \rfloor \\ \text{其他 } k = \dots \end{cases}$

2) 零点: $\sin(N\beta) = 0$ 且 $\sin\beta \neq 0$

$$\beta = (k + \frac{m}{N})\pi \quad m \in [1, N) \quad \text{有 } N-1 \text{ 个}$$

$$\text{即 } \sin\theta = (k + \frac{m}{N}) \frac{\lambda}{d} \quad [\text{次极大在零点间, 有 } N-2 \text{ 个}]$$

→ 规定: 主极大半角宽度 = 主极大中心到相邻暗纹的角宽度

$$\sin\theta_k = \frac{k\lambda}{d}, \quad \sin(\theta_k + \Delta\theta_k) = (k + \frac{1}{N}) \frac{\lambda}{d}$$

$$\Delta\theta_k \approx \frac{k\lambda}{d \cos\theta_k}$$

5.4 单缝衍射图样

$$\text{零点: } \alpha = n\pi \rightarrow \sin\theta_n = \frac{n\lambda}{a}$$

若某 θ_n 与 θ_k 重合, 则出现“缺级”

5.5 黑白光栅和正弦光栅 (一维周期性结构的衍射)

在波前之上 $\tilde{u}(x)$ 是周期函数

$$L_2 = L_1 + \Delta L, \quad \dots \quad L_n = L_1 + (n-1)\Delta L$$

$$\text{其中 } \Delta L = d \sin\theta$$

$$\begin{aligned} \tilde{U}(\theta) &= C \int_{-\infty}^{\infty} \tilde{u}_0(x_j) e^{ikr_j} dx_j \\ &= \sum_{j=1}^N C \int_{-\infty}^{\infty} \tilde{u}_0(x_j) e^{ikr_j} dx_j \end{aligned}$$

$$\text{其中 } r_j = L_j - x_j \sin\theta$$

解得:

第 j 个结构 $\begin{matrix} x \\ \uparrow \\ 0 \end{matrix} \begin{matrix} \nearrow \theta \\ \cdot \\ \searrow \end{matrix} \begin{matrix} L_j \\ \downarrow \\ 0 \end{matrix}$

$$\int \tilde{u}_0(x_j) e^{ikr_j} dx_j = e^{ikL_j} \int \tilde{u}_0(x_j) e^{-ikx_j \sin\theta} dx_j$$

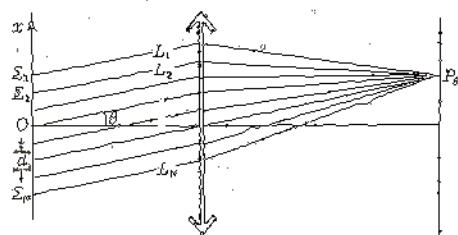


图 4-42 一维周期性结构的衍射

$$= e^{ikL_j} \int_{-\frac{d}{2}}^{\frac{d}{2}} \tilde{u}_0(x) e^{-ikx \sin \theta} dx$$

$$\begin{aligned} \tilde{U}(\theta) &= \sum_j c_j \cdot e^{ikL_j} \int_{-\frac{d}{2}}^{\frac{d}{2}} \tilde{u}_0(x) e^{-ikx \sin \theta} dx \\ &= \left[\sum_j e^{ikL_j} \right] \cdot \underbrace{c_j \cdot \int_{-\frac{d}{2}}^{\frac{d}{2}} \tilde{u}_0(x) e^{-ikx \sin \theta} dx}_{N(\theta)} = \tilde{N}(\theta) \tilde{u}_0(\theta) \end{aligned}$$

元衍射图 $N(\theta)$ 元衍射图 $\tilde{u}_0(\theta)$

$$\tilde{u}(\theta) = c \cdot \int_{-\frac{d}{2}}^{\frac{d}{2}} \tilde{u}_0(x) e^{-ikx \sin \theta} dx$$

$$\begin{aligned} \tilde{N}(\theta) &= e^{ikL_1} [1 + e^{ik\alpha L} + \dots + e^{ik(N-1)\alpha L}] \\ &= e^{ikL_1} \frac{1 - e^{ikN\alpha L}}{1 - e^{ik\alpha L}} \end{aligned}$$

$$\begin{aligned} \lambda \sin \theta &= \frac{2\pi d}{\lambda} \sin \theta \\ &= e^{ikL_1} \frac{1 - e^{2iN\pi}}{1 - e^{2i\pi}} = e^{ikL_1} \cdot \frac{e^{iN\pi} - e^{-iN\pi}}{e^{i\pi} - e^{-i\pi}} \\ &= e^{i[kL_1 + (N-1)\pi]} \cdot \frac{\sin(N\pi)}{\sin(\pi)} \end{aligned}$$

$$\text{记 } \varphi = kL_1 + (N-1)\pi \quad N(\theta) = \frac{\sin N\pi}{\sin \pi}$$

$$\tilde{N}(\theta) = e^{i\varphi} N(\theta)$$

在上式中

$$\textcircled{1} \text{ 代入单缝衍射时 } \tilde{u}_0(x) = \begin{cases} \text{const.} & x \in [-\frac{d}{2}, \frac{d}{2}] \\ 0 & x \notin [-\frac{d}{2}, \frac{d}{2}] \end{cases}$$

(矩形) 函数

$$\begin{aligned} \text{可得 } \tilde{u}(\theta) &= \int_{-\frac{d}{2}}^{\frac{d}{2}} e^{-ikx \sin \theta} dx = \frac{1}{ik \sin \theta} (e^{ik \sin \theta \cdot \frac{d}{2}} - e^{-ik \sin \theta \cdot \frac{d}{2}}) \\ &= \frac{\sin \alpha}{\alpha} \quad \text{其中 } \alpha = \frac{2\pi d \sin \theta}{\lambda} \end{aligned}$$

$$\text{此时 } \tilde{U}(\theta) = c \cdot e^{i\varphi} \left[\frac{\sin N\pi}{\sin \pi} \right] \cdot \left(\frac{\sin \alpha}{\alpha} \right)$$

$$\textcircled{2} \text{ 代入正弦型光栅时 } \tilde{u}_0(x) = \delta(1 + \cos \frac{2\pi x}{\Lambda})$$

$$\begin{aligned}\tilde{u}(\theta) &= C \int_{-\frac{a}{2}}^{\frac{a}{2}} \left[1 + \cos\left(\frac{2\pi x}{d}\right) \right] e^{-ikx \sin \theta} dx \\ &= C' \left[\int_{-\frac{a}{2}}^{\frac{a}{2}} e^{-ikx \sin \theta} dx + \int_{-\frac{a}{2}}^{\frac{a}{2}} \cos\left(\frac{2\pi x}{d}\right) e^{-ikx \sin \theta} dx \right] \\ &\quad C' \quad (I_1 \quad + \quad I_2)\end{aligned}$$

$$I_1 = \text{sinc}\left(\frac{ka \sin \theta}{2}\right) = \text{sinc}\left(\frac{\lambda a \sin \theta}{\lambda}\right)$$

$$\begin{aligned}I_2 &= \int_{-\frac{a}{2}}^{\frac{a}{2}} \frac{1}{2} (e^{i\frac{2\pi x}{d}} + e^{-i\frac{2\pi x}{d}}) e^{-ikx \sin \theta} dx \\ &= \frac{1}{2} \left[\int_{-\frac{a}{2}}^{\frac{a}{2}} e^{-i(k \sin \theta - \frac{2\pi}{d})x} dx + \int_{-\frac{a}{2}}^{\frac{a}{2}} e^{-i(k \sin \theta + \frac{2\pi}{d})x} dx \right] \\ &= \frac{1}{2} \left\{ \text{sinc}\left[\frac{a}{2}(k \sin \theta - \frac{2\pi}{d})\right] + \text{sinc}\left[\frac{a}{2}(k \sin \theta + \frac{2\pi}{d})\right] \right\} \\ &= \frac{1}{2} \left[\text{sinc}\left(\frac{\lambda a \sin \theta}{\lambda} - \lambda\right) + \text{sinc}\left(\frac{\lambda a \sin \theta}{\lambda} + \lambda\right) \right]\end{aligned}$$

$$\rightarrow \tilde{u}(\theta) = C' \left\{ \text{sinc} \beta + \frac{1}{2} [\text{sinc}(\beta + \lambda) + \text{sinc}(\beta - \lambda)] \right\}$$

$$U(\theta) = C' e^{i\varphi} \frac{\text{sinc}(N\beta)}{\text{sinc} \beta}$$

$$\left\{ \text{sinc} \beta + \frac{1}{2} [\text{sinc}(\beta + \lambda) + \text{sinc}(\beta - \lambda)] \right\}$$

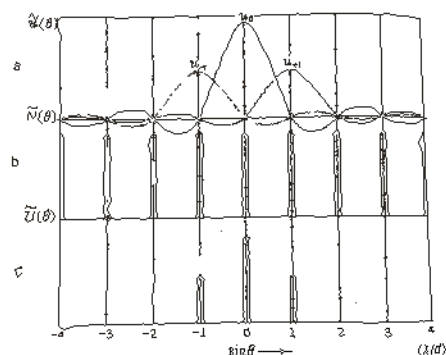


图 4-44 正施光栅的振幅分布

§ 6. 光栅光谱仪

1. 光栅分光原理

光栅公式: $\sin \theta = k \frac{\lambda}{d}$ d 称为光栅常量

2. 光栅的色散本领和色分辨本领

1) 定义: 色散本领 D_0/D_1

“对一定波长差 $\Delta \lambda$ 的两条谱线, 距离有多大”

角色散本领 $D_0 = \frac{d\theta}{d\lambda}$ 线色散本领 $D_1 = \frac{\delta l}{d\lambda}$

$$\Delta l = f \cdot \Delta \theta \rightarrow D_l = f \cdot D_\theta$$

对光栅公式两端取微分：

$$\cos \theta_k \Delta \theta_k = \frac{k \Delta \lambda}{d}$$

$$\text{得： } D_\theta = \frac{k}{d \cos \theta_k} \quad D_l = \frac{f k}{d \cos \theta_k}$$

例：Na 的两条谱线 $\lambda = 589.00 \text{ nm}$ 和 $\lambda' = 589.59 \text{ nm}$ 。

使用 15 cm ，每毫米内刻有 1200 条缝的光栅

1 级光谱中两谱线位置、间隔、半角宽度为多少？

$$\text{位置： } \sin \theta_{11} = \frac{\lambda}{d}, \quad \sin \theta_{12} = \frac{\lambda'}{d} \quad \text{其中 } d = \frac{1}{1200} \text{ mm}$$

$$\text{间隔： } \sin \theta_{12} = \sin(\theta_{11} + \Delta \theta) = \sin \theta_{11} \cos \Delta \theta + \cos \theta_{11} \sin \Delta \theta$$

$$\approx \sin \theta_{11} + \cos \theta_{11} \Delta \theta$$

$$\Rightarrow \Delta \theta = \frac{\sin \theta_{12} - \sin \theta_{11}}{\cos \theta_{11}}$$

$$\text{半角宽： } \beta \approx (1 + \frac{1}{N}) \gamma = \frac{N d (\sin \theta_{11} + \Delta \theta_{11})}{\lambda}$$

$$\Delta \theta_{11} = \frac{\lambda}{N d \cos \theta_{11}}$$

$$\Delta \theta_{12} = \frac{\lambda}{N d \cos \theta_{12}}$$

(2) 定义：色分辨率 R

瑞利判据：谱线 λ 和 $\lambda' = \lambda + \Delta \lambda$ 的角距离 $\Delta \theta$ 恰等于半角宽度 $\Delta \theta$ 的

认为“可分辨”

最小可分辨的波长差：

$$\Delta \lambda = \frac{\Delta \theta}{D_\theta} = \frac{\Delta \theta}{\frac{k}{d \cos \theta_k}} = \frac{\lambda}{k N}$$

$$R := \frac{\lambda}{\delta\lambda} = kN$$

例: 15cm 宽的光栅, 每 mm 内 1200 个单元

在 $\lambda = 550 \text{ nm}$ 波长的分辨率最小波长差?

$$\delta\lambda = \frac{\lambda}{R} = \frac{\lambda}{D_0} = \frac{\lambda}{kN} \quad \text{取 } k=1$$

$$(\delta\lambda)_{\min} = \frac{\lambda}{N} = \frac{\lambda d}{L}$$

∴ 量程和自由光谱范围

$$\sin\theta \leq 1 \Rightarrow \lambda_{\max} < d$$

一般 $\lambda_{\min} > \frac{1}{2} \lambda_{\max}$ (避免相邻光谱重叠)

4. 闪耀光栅

闪耀角 θ_b

a. 沿 n 射入

槽间干涉:

$$\Delta L = 2d \sin\theta_b = k \lambda_{kb}$$

$$k \text{ 级闪耀波长 } \lambda_{kb} = \frac{2d \sin\theta_b}{k}$$

单槽衍射: 0 级 (重合) 槽间干涉: 1 级

因单槽衍射 0 级主峰有一定宽度

⇒ λ_b 附近 ±1 级谱线亦有较大亮度

b. 沿 N 射入

$$\gamma = \gamma_1 \sin^2 \alpha + \gamma_2 \sin^2 \beta \quad \alpha = \frac{2\lambda d \sin\theta}{\lambda} \quad \gamma_2 = \frac{2\lambda d \sin\theta}{\lambda}$$

$$\frac{1}{2} \frac{d}{dt} \left(\frac{1}{\sin^2 \beta} \right) = \frac{1}{\lambda} \frac{d\lambda}{dt}$$

$$\Delta L = d \sin(2\theta_b) = \lambda_{1b}'$$

§ 7. 二维光栅

2. 布拉格条件

1. 点阵干涉 — 同一平面内

0级主极大方向?

各衍射线 $\Delta L = 0$ 方向

$\rightarrow \theta' = \theta$ (另有 $\theta' = -\theta$)

2. 缝间干涉

$$2d \sin \theta = k\lambda$$

与光栅 $d \sin \theta = k\lambda$ 的不同:

① 一个晶体内有多个晶面族, 对应多个 θ

② 在光栅中, θ 是衍射角, 任意取值

而晶体中 θ 已给定

